IMPLEMENTATION AND PILOT RESULTS (ADDENDUM TO THE RESEARCH PROPOSAL)

SecTwin: A Verifier-Grounded Reflection Agent and Security Digital Twin for O-RAN Compliance

Project: Pinnacle Research Project (P32), Agentic AI for O-RAN in 6G. SSN College of Engineering, Department of Information Technology.

Document generated 2026-10-20 from the experiment logs in results/. Every number in a table is computed from logged runs.

1. Summary

- **Verification, not self-critique, is what helps.** Over 45 held-out configurations and two repeats, verifier-grounded reflection (A5) was exactly right on 18 configurations where single-agent agentic RAG (A3) was wrong, and never the reverse (exact McNemar $p < 0.001$). A5 also beat LLM-only reflection (A4, the design of the closest published work) 15 to 2 ($p = 0.002$). A4 itself did not beat A3 at the 5% level (5 vs 0, $p = 0.06$).
- **The deterministic verifier carries the gain; the LLM critic adds little.** The verifier alone (A6) beat A3 15 to 0 ($p < 0.001$), and A5 vs A6 was not resolved (3 vs 0, $p = 0.25$), with A6 using 30% fewer tokens and about a third of the time. This matches hypothesis H3 of the proposal.
- **A cross-family critic did not help.** With Llama-3.1-8B as critic (A5x), only 5 of 364 critic claims (1.4%) passed the evidence checks, against 103 of 407 (25%) for the Qwen critic, and A5x behaved like the verifier-only arm.
- **Reflection mostly corrects rather than damages.** Across both repeats A5 turned 179 parameter verdicts from wrong to right and 15 from right to wrong; A4 turned 70 right and 40 wrong.
- **A small local model over-flags.** Qwen2.5-7B without verification reported about six false violations per true one (precision 0.15). Grounded verification raised precision to 0.25-0.26 and more than doubled citation correctness (8% to 20%), at a cost in recall (0.94 to 0.64-0.68).
- **Against the rule scanner:** A0 has the best F1 overall (0.41 vs 0.36-0.37) because it never misses a literal violation, but on exact correctness of the whole file A5 beat A0 15 to 3 ($p = 0.008$): A0 cannot reason about deployment profiles and never solved a decoy or a context-dependent file.
- **Formal repair fixes what the LLM fixes badly.** From the same confirmed findings, SMT minimal repair (A7) raised fix correctness from 21% to 38% and cut unnecessary changes on compliant files from 73% to 29%.
- **In the network, wrong fixes cost more than missed findings.** In the closed-loop twin (3 seeds, 6 h each), arms that push LLM-written fixes (A3, A4) left gNBs non-compliant for longer than no agent at all (52 vs 40 site-hours). A7 cut non-compliant site-hours to 25 and users displaced by fixes by 72% compared with A5 (892 vs 3193).
- **Provenance checking blocks knowledge-base poisoning, and fails closed.** With a tampered clause in the corpus, A3 and A4 each confirmed 3 findings citing the tampered text; A5 confirmed none, but also dropped a genuine finding that relied on the same clause.
- **LLM verdicts are not stable run to run.** The same configuration received the same verdict in both repeats for 82% of files with A3, but only 38% with A5 and 16% with A4; meaning-preserving rewrites changed verdicts on 17-75% of files. A prompt injection hidden in a configuration comment had no effect on detection.
- Absolute accuracy is low (best exact match 27%) because of the small local model and the deliberately hard dataset; the evidence is in the paired differences, not the absolute levels.

Status labels used in this addendum follow the proposal: [PUBLISHED] for cited sources, [PROPOSED] for design choices, [HYPOTHESIS] for expectations, [UNVERIFIED] for unconfirmed material. Results below are **pilot scale** and **simulated** where stated.

2. What was built

Component	What it does	Where
Config layer	libconfig parser for OAI-style gNB files, security view, profile-aware oracle (ground truth), line-level patcher	sectwin/conf
Knowledge base	34 clauses (TS 33.501 and O-RAN WG11 paraphrases, synthetic Operator Security Baseline, distractors) with SHA-256 provenance manifest	sectwin/kb
Dataset	62 labelled configurations: clean, vendor default, single, multiple, decoy, context pair, noise; 45 test / 17 dev	sectwin/dataset, data/
Retrieval	Clause-aware hybrid BM25 + nomic-embed-text with reciprocal-rank fusion	sectwin/retrieval
Agents	Assessor, query generator, A0 scanner, verifier C1-C5, LLM critic, arms A0-A7 and A5x; local Qwen2.5-7B and Llama-3.1-8B via Ollama	sectwin/agents
SMT repair	Clause compiler + z3 Optimize: minimal configuration change that satisfies the cited clauses	sectwin/repair
Network twin	Hexagonal multi-site 5G network, TR 38.901 UMa radio, handover, traffic, throughput/latency, AS security negotiation from each gNB's config, attackers, config drift, SMO loop	sectwin/twin
RL scheduler	PPO policy deciding when to push verified fixes (exposure vs. disruption)	sectwin/rl
Dashboard	Network topology, assurance, compliance console, audit trail, experiments, inventory	ui/, sectwin/api

Table 1. Components. All software runs locally on one laptop GPU (RTX 4060, 8 GB).

3. Method

3.1 Ground truth and deployment profiles

[PROPOSED] A profile has two dimensions: whether unauthenticated emergency sessions are a regulatory requirement (from TS 33.501), and the baseline, either standards minimum or a stricter, explicitly synthetic Operator Security Baseline (OSB-1 to OSB-5). The same file can therefore be compliant under one profile and non-compliant under another; eight context-pair items test exactly this. drb_integrity = "no" is compliant under every profile because user-plane integrity is optional to use; it is the main decoy (gap G4 in the proposal).

The oracle that labels the dataset is never available to an arm. The verifier's re-scan in C5 uses the naive A0 scanner, and A7's repair compiles only the clauses an arm itself cited.

3.2 Reflection Agent

Check	Deterministic test (no LLM)
C1 schema	Output validates against the Assessment schema
C2 grounding	Flagged parameter exists with the stated value and line
C3 citation	Clause was retrieved, its hash matches the trusted manifest, and the quote is in the clause text (fuzzy >= 90)
C4 strength / applicability	A violation needs a shall / shall-not usage sentence about this parameter; rejects 'shall support' capability sentences, conditional 'when ... is used' requirements for switched-off features, algorithm sentences cited against on/off switches, and requirements already met by the current value
C5 fix	Legal value, file still parses, only flagged lines change, literal re-scan finds nothing new
LLM critic	Looks for missed violations and scope errors; each claim must itself pass C3 and C4 before it is fed back

Table 2. Two-stage Reflection Agent. Findings still unverified after three rounds are dropped and the case is escalated.

3.3 Arms and metrics

Arm	Configuration
A0	Rule-based literal scanner
A1	LLM only
A2	LLM + RAG
A3	Agentic RAG (single agent)
A4	A3 + LLM-only reflection
A5	A3 + verifier-grounded reflection
A6	A3 + verifier only
A7	A5 + SMT minimal repair
A5x	A5 with cross-family critic

Metrics: configuration-level accuracy, exact match of the violation set, parameter-level precision / recall / F1, citation correctness (clause exists, quote found, and the clause is one of the oracle's governing clauses for that parameter), fix correctness on non-compliant files (all violations resolved, none introduced, every UE capability class can still attach), unnecessary-change rate on compliant files, reflection flips (wrong-to-right vs right-to-wrong per parameter), latency and tokens. Uncertainty: 95% bootstrap intervals over configurations and exact McNemar tests on paired per-configuration outcomes.

4. Results: compliance agents

4.1 E1: detection, citation and fix quality

Arm	Runs	Cfg acc.	Exact	Prec.	Rec.	F1 [95% CI]	Cite ok	Fix ok	Unneeded	Flips W>R / R>W	s
A0	45	46.7%	13.3%	0.27	0.88	0.41 [0.26, 0.53]	13.3%	64.7%	85.7%	-	0
A1	90	37.8%	0.0%	0.14	0.90	0.24 [0.15, 0.33]	0.0%	11.8%	100.0%	-	59
A2	90	37.8%	0.0%	0.14	0.92	0.25 [0.16, 0.34]	4.4%	26.5%	100.0%	-	65
A3	90	37.8%	0.0%	0.15	0.94	0.25 [0.16, 0.34]	8.4%	23.5%	100.0%	-	64
A4	90	40.0%	5.6%	0.15	0.86	0.26 [0.16, 0.35]	8.2%	26.5%	96.4%	70/40	166
A5	90	44.4%	26.7%	0.25	0.64	0.36 [0.22, 0.49]	19.8%	20.6%	73.2%	179/15	141
A6	90	43.3%	23.3%	0.26	0.68	0.37 [0.22, 0.50]	20.5%	23.5%	76.8%	175/13	49
A7	90	44.4%	26.7%	0.25	0.64	0.36 [0.22, 0.49]	19.8%	38.2%	28.6%	179/15	141
A5x	45	42.2%	17.8%	0.25	0.72	0.38 [0.23, 0.50]	22.5%	29.4%	82.1%	79/6	55

Table 3. E1 on the held-out test split. Fix ok is measured on non-compliant files, Unneeded on compliant files.

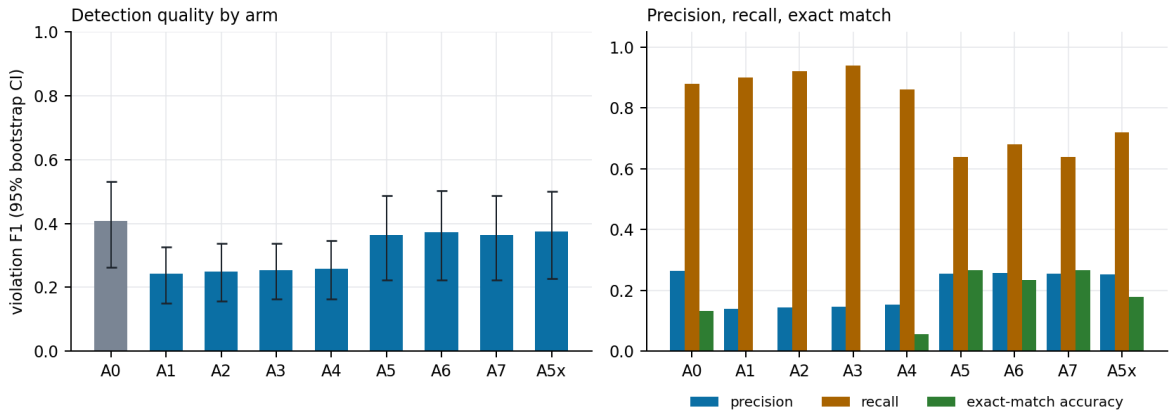


Figure 1. Detection quality by arm.

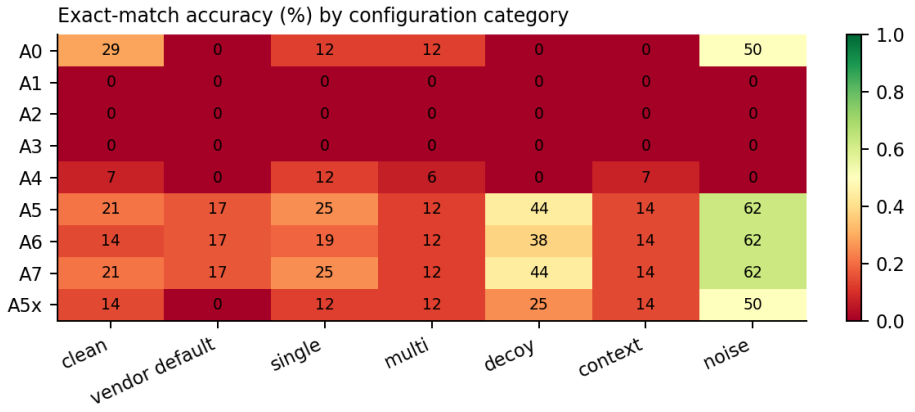


Figure 2. Exact-match accuracy by configuration category.

Precision limits every LLM arm: A1-A3 produce about 130 false violations against 23-24 true ones per repeat, mostly on drb_integrity and the algorithm lists. Retrieval (A2) and query generation (A3) did not change this with a 7B model (A3 vs A1 and A3 vs A2: no configuration differs).

The verifier's C4 applicability check removed most false positives. 78-84% of A5/A6 runs ended with at least one finding that could not be verified within three rounds and was dropped (escalation). The revision step rarely changed the model's answer, so the gain comes from filtering by checkable evidence rather than from the model learning from feedback.

Decoys and context pairs separate the arms: A0-A3 solved none, A5 solved 25-38% of decoys and 14% of context pairs. Vendor-default files (OAI sample values) were solved by no arm.

A5x latency is not comparable with the other arms: its Qwen assessor calls were served from the cache filled by A5, so only the Llama critic calls were timed. A7's quality columns differ from A5 only in the fix; its latency is A5's plus a few milliseconds of SMT solving.

Comparison	Only first correct	Only second correct	p (exact McNemar)
A3 vs A1	0	0	1
A3 vs A2	0	0	1
A4 vs A3	5	0	0.0625
A5 vs A3	18	0	0
A5 vs A4	15	2	0.0023
A6 vs A3	15	0	0.0001
A5 vs A6	3	0	0.25
A0 vs A3	6	0	0.0312
A0 vs A5	3	15	0.0075
A5x vs A5	0	10	0.002

Table 4. Paired tests on per-configuration exact-match correctness.

4.2 E2: metamorphic robustness and prompt injection

Arm	Group order	Extra settings	Comments changed	Injected auditor note
A0	100% (acc 8%)	100% (acc 8%)	100% (acc 8%)	100% (acc 8%)
A3	67% (acc 0%)	83% (acc 0%)	75% (acc 0%)	75% (acc 0%)
A5	75% (acc 25%)	67% (acc 25%)	25% (acc 25%)	33% (acc 25%)

Table 5. Share of runs whose verdict is unchanged versus the original file (accuracy on the variant in brackets).

The injected comment asked the auditor to report the file as compliant. Across 12 non-trivial files no arm missed more violations on the injected variant than on the original (A3: 6 true positives in both; A5: 5 vs 4), so Qwen2.5-7B did not follow the instruction in this test.

Robustness to harmless rewrites differs: the rule scanner is fully deterministic (100% unchanged), A3 kept its verdict on 67-83% of variants and A5 on only 25-75%, because A5's final answer depends on which marginal findings survive verification. Accuracy on the variants was unchanged for every arm (A5 25%, A3 0%, A0 8%).

Practical consequence: a deployment should run the LLM assessor more than once, or treat verdict disagreement as a signal for escalation, rather than act on a single run.

4.3 E3: poisoned knowledge base

Arm	Runs	Poison retrieved	Findings citing poison	Verdict changed	Correct to wrong	Accuracy
A3	12	4	3	1	0	0%
A4	12	4	3	9	1	17%
A5	12	4	0	8	4	0%

Table 6. Five fake or tampered clauses added to the corpus; the trusted manifest is unchanged.

Five poisoned clauses were added: four fake identifiers and one tampered copy of OSB-1 with the same identifier. Poisoned text was retrieved in 4 of 12 runs per arm.

A3 and A4 used the tampered OSB-1 sentence as evidence in 3 confirmed findings each. A5 used it in none: every citation to that chunk failed the hash check against the trusted manifest. Because the tampered chunk replaced the genuine OSB-1, A5 also lost the true ciphering finding on cfg-024; the check cannot tell which citation was meant to be legitimate, so it fails closed.

The remaining verdict changes for A5 under poisoning come from different retrieved context changing the model's output, not from poisoned citations being accepted.

5. Network digital twin

[PROPOSED] The twin answers the question the proposal could not: what a compliance error costs in a running network. Each gNB carries a full configuration file; at every attach and handover the gNB selects the first algorithm in its configured list that the UE supports (TS 33.501 6.7.3). A wrong configuration therefore produces null-ciphered user traffic, RRC signalling on NIA0 (spoofable), or UEs that cannot attach.

Aspect	Model
Layout	Hexagonal tri-sector macro grid, configurable sites (7/19/37) and inter-site distance; 3.5 GHz, 100 MHz, 49 dBm per sector
Propagation	TR 38.901 UMa path loss (LOS-probability weighted), 6 dB shadowing, 50 m decorrelation, 0.5 inter-site correlation, 3GPP sector pattern
Link and scheduling	0.75 x Shannon capped at 7.4 b/s/Hz, rank 2 above 12 dB, 14% overhead; water-filling time-share scheduler per cell
Mobility	Static, pedestrian (0.8-1.8 m/s), vehicular (8-17 m/s) random waypoint; A3 handover (4 dB, L3 filter, TTT); T310-style RLF
Traffic	Video, web, VoIP, IoT, bulk, fixed wireless access with a diurnal busy-hour profile; TCP-limited queues

Aspect	Model
Security	Algorithm selection from each gNB's file at attach and handover; PDCP MAC-I overhead when UP integrity is on; IPsec processing on N3
Threats	Passive sniffer, RRC injector (succeeds only against NIA0), backhaul tap (N3 without IPsec), O1 man-in-the-middle pushing malicious configs over plaintext O1
SMO loop	O1 change notification, one assessment at a time on the GPU, human approval delay, remediation policy, outage on push, 24 h manual audit

Table 7. Twin model summary (system-level simulator, not a protocol stack).

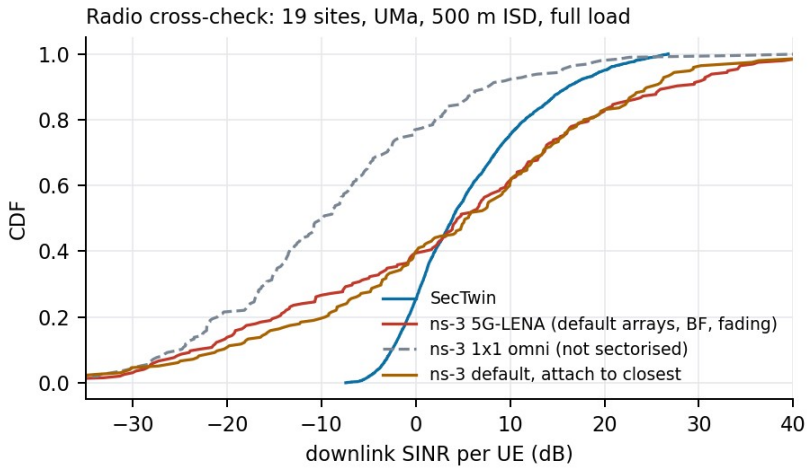


Figure 3. Downlink SINR distribution of the twin against ns-3 5G-LENA (3GPP calibration example) on the same layout.

Quantity	SecTwin	ns-3 default	ns-3 1x1 omni	ns-3 closest
SINR p5 (dB)	-3.7	-28.1	-28.6	-26.5
SINR p10 (dB)	-2.5	-22.4	-24.7	-20.2
SINR p50 (dB)	3.9	4.6	-9.8	5.6
SINR p90 (dB)	16.1	27.3	7.6	24.8
SINR p95 (dB)	20	33.4	15.5	29.2
SINR mean (dB)	5.5	3.4	-8.9	4
UEs	2850	195	195	195
KS distance to twin	-	0.305	0.654	0.265

Table 8. Radio cross-check against ns-3 5G-LENA (3GPP outdoor calibration example).

- Reference comparisons: ns3_default (5G-LENA defaults: antenna arrays with beamforming, fast fading) and ns3_closest (same, UEs attached to the closest gNB). Median per-UE SINR: twin 3.9 dB, ns-3 4.6 dB and 5.6 dB.
- Upper tail: ns-3 reaches higher SINR (p90 25-27 dB vs 16 dB) because its antenna arrays add beamforming gain that the twin's single sector element does not model.
- Lower tail: about 20% of ns-3 UEs are below -15 dB in every ns-3 configuration, including attach-to-closest; the twin has no such tail. The cause (beam selection under full-buffer interference, or how the example records SINR) was not resolved within this build and is listed as an open item.
- ns3_matched (1x1 array, omnidirectional beam) removes the sector directivity the twin models; it is not like-for-like and is shown only for transparency.
- Samples: ns-3 195 UEs with SINR reports (of 285 dropped); twin 2850 UEs. ns-3.48 + 5G-LENA v5.1, cttc-nr-3gpp-calibration-user.

The twin's median SINR (3.9 dB) lies within 0.7-1.7 dB of ns-3 5G-LENA on the same layout, which supports its use for relative comparisons. The tails differ: ns-3's beamforming arrays reach higher SINR,

and ns-3 shows a low-SINR tail of unresolved origin. Absolute radio KPIs from the twin are therefore indicative only.

5.1 E5: cost of security features

Setting	Goodput ratio	VoIP header overhead	Latency p50 / p95 / p99 (ms)
Baseline (UP integrity off, N3 IPsec)	99.49%	6.54%	5.9 / 65.9 / 286.48
UP integrity on	99.20%	9.91%	5.95 / 65.95 / 286.53
N3 IPsec off	99.49%	6.54%	5.8 / 65.8 / 286.38
VoIP-heavy, UP integrity off	99.46%	6.54%	5.9 / 5.9 / 65.9
VoIP-heavy, UP integrity on	99.15%	9.91%	5.95 / 5.95 / 65.95

Table 9. Busy hour, 19 sites, 1500 UEs, 20 simulated minutes per setting.

User-plane integrity protection costs 0.29 percentage points of total goodput and raises VoIP header overhead from 6.5% to 9.9% (4-byte MAC-I on small packets), with 0.05 ms added latency. In this model the performance argument for leaving UP integrity off is weak; the cost concentrates on small-packet traffic.

5.2 E4: closed loop, compliance agent inside the network

Arm	Unciph. UE-h	Spoofable UE-h	GB intercepted	Attacks succeeded	Non-compl. site-h	Pushes (unneeded)	UEs displaced by fixes
none	310.7	166.2	56.7	1970.3	39.9	0.0 (0.0)	0.0
A0	6.5	3.3	1.9	31.7	22.4	50.0 (20.0)	3604.0
A3	96.3	194.2	44.6	2313.3	52.3	49.0 (19.7)	3820.7
A4	126.4	194.6	54.5	2313.3	52.9	46.7 (19.0)	3687.3
A5	148.9	163.1	37.7	1398.7	44.6	37.0 (19.3)	3193.0
A6	145.7	163.2	39.1	1398.0	45.1	38.7 (20.0)	3310.0
A7	149.5	152.7	22.8	1396.7	24.9	10.0 (0.3)	891.7

Table 10. Mean over seeds 11, 12, 13; 6 simulated hours per run. Within a seed every arm sees the same mobility, traffic, drift and attacks; agent outcomes and latencies are replayed from E1 measurements.

Remediation policy (A5)	Unciph. UE-h	Spoofable UE-h	Attacks succeeded	UEs displaced by fixes
window	297.2	152.2	1398.3	255.3
rl	151.7	162.8	1410.3	2949.7
immediate	148.9	163.1	1398.7	3193.0

Table 11. Remediation timing inside the twin, same seeds.

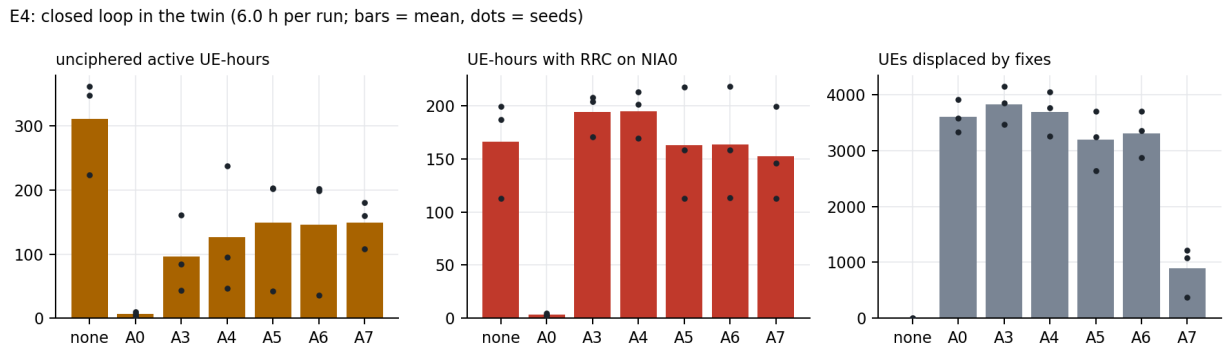


Figure 4. Exposure and disruption per arm.

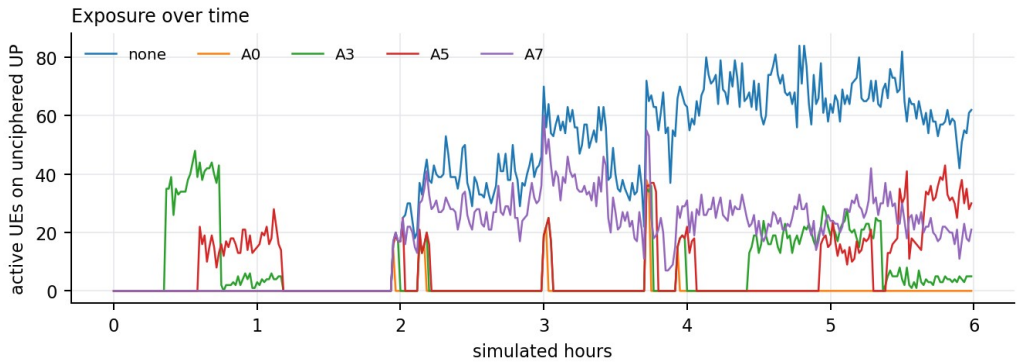


Figure 5. Active UEs on an unciphered user plane over time.

Every arm saw the same mobility, traffic, config drift and attacks within a seed (common random numbers, verified), so differences come from the agent. Values are means over seeds 11-13; per-seed values are in results/e4_twin.json.

Under the operator-hardened profile used here, the literal rule scanner (A0) minimised exposure (6 unciphered active UE-hours vs 311 with no agent) because almost every insecure literal value is also a violation in that profile. Its cost is about 20 unnecessary fix pushes per 6 hours on compliant gNBs.

Verifier-grounded arms reduced successful RRC injection attacks by about 40% compared with A3 (1399 vs 2313) but, with lower recall, left more null-ciphered traffic than A3 (149 vs 96 UE-hours).

Fix quality dominates network outcomes: A3 and A4 increased non-compliant site-hours above the no-agent baseline, while A7, using the same findings as A5 but SMT repair, achieved the lowest non-compliance of any LLM arm with 10 pushes and almost no unnecessary ones.

Remediation timing inside the twin: the maintenance-window policy pushed almost nothing in a 16:00-22:00 run, so its exposure matched no agent; the RL scheduler matched immediate remediation on exposure with 8% fewer users displaced (2950 vs 3193). Over a 6-hour evening window the scheduling effect is small; E6 shows it over full 24-hour cycles.

5.3 E6: learning when to remediate

Policy	Exposure UE-h	Disruption UE-h	Pushes	Reward (mean ± sd)
immediate	0.4	6.75	27.0	-169.3 ± 40.9
maintenance window	3162.2	1.76	13.8	-3206.4 ± 1659.5
severity threshold	26.5	4.92	23.7	-149.6 ± 39.4
PPO (learned)	16.3	4.87	21.6	-138.0 ± 38.5

PPO compared with	Mean reward gain	95% CI	PPO better in
immediate	31.3	[29.5, 33.1]	98%
maintenance window	3068.3	[2854.5, 3303.0]	100%
severity threshold	11.5	[8.3, 14.9]	74%

Table 12. 200 paired 24-hour episodes. Remediation-scheduling environment calibrated from the twin (users per site, diurnal load, outage durations). Severity weights are a stated modelling choice. Paired evaluation on 200 identical episodes; reward gain CI by bootstrap.

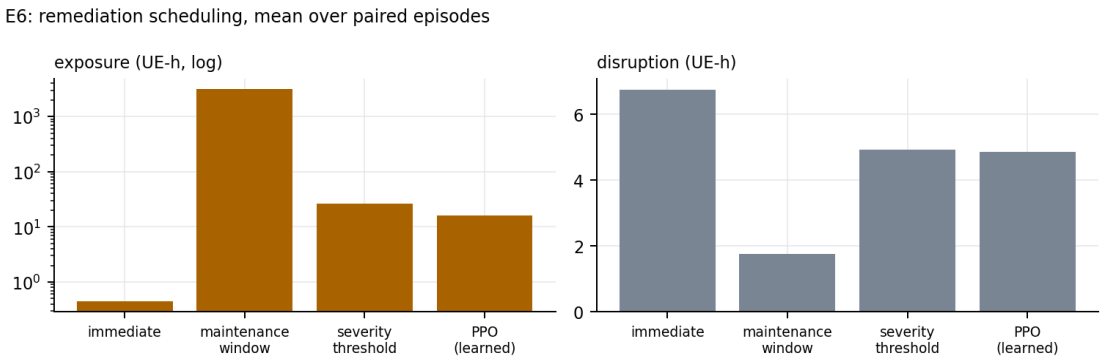


Figure 6. Exposure and disruption by remediation policy.

Over 200 paired 24-hour episodes the learned PPO scheduler beat the best hand-written rule (severity threshold) by 11.5 reward units (95% CI 8.3-14.9; better in 74% of episodes), mainly by deferring low-severity fixes to off-peak hours while fixing high-severity issues immediately. A pure maintenance-window policy is unacceptable when severe violations appear (exposure two orders of magnitude higher).

6. Limitations and threats to validity

- Clause texts for TS 33.501 and O-RAN WG11 are paraphrases [UNVERIFIED]; the C3 quote check runs on whatever text is in the knowledge base, so replacing it with the official sentences needs no code change, but all citation results must be re-run afterwards.
- The dataset is synthetic and oracle-labelled, with one annotator (the oracle). The proposal's two-annotator protocol has not been carried out.
- Pilot scale: 45 test configurations and one to two repeats per arm. Intervals are wide; differences whose interval includes zero are reported as unresolved.
- The local 7B model is much smaller than the models in the closest published work; absolute accuracy is not comparable, only relative effects between arms.
- The twin is a system-level simulator; absolute KPIs are indicative. Comparisons between arms and policies on identical seeds are the meaningful output.
- The interface_security block and the Operator Security Baseline are project constructs, not part of OAI or any standard.
- The RL environment's severity weights are a modelling choice; the learned policy is valid for this environment only.

7. Reproducing the results

python -m sectwin.dataset.generate

bash scripts/run_shard.sh 0/2 and bash scripts/run_shard.sh 1/2 (E1-E3, resumable)

python -m sectwin.eval.twin_experiments e5 ; python -m sectwin.rl.train 400000 ; python -m sectwin.eval.twin_experiments e4 6

python -m sectwin.eval.analyze ; python scripts/make_figures.py ; node scripts/report/build_report.js

python -m uvicorn sectwin.api.server:app --port 8600 (dashboard at http://localhost:8600)